METRIC

MIL-P-62424(AT)
16 August 1991
SUPERSEDING
~~(see 6.9)~~

# MILITARY SPECIFICATION

## POWDER, YTTRIUM, ALUMINUM, NICKEL-CHROMIUM (METRIC)

This specification is approved for use by the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a nickel-chromium, aluminum, yttrium (94Ni-Cr, 6Al, 1.0Y) powder used in plasma spray applications, hereinafter referred to as powder (See 6.1).

1.2 Classification. The powder shall come in the following unit container sizes (see 3.3 and 6.2).

1.2.1 Metric unit container sizes.

| Unit Container Size | Part Identification Number (PIN) |
|---|---|
| 0.5 Kilogram (Kg) | MS62424-0.5 Kg |
| 2.5 Kg | MS62424-2.5 Kg |
| 10.0 Kg | MS62424-10.0 Kg |

1.2.2 English unit container sizes.

| Unit Container Size | PIN |
|---|---|
| 1.0 pound (lb) | MS62424-1.0 lb |
| 5.0 lbs | MS62424-5.0 lbs |
| 25.0 lbs | MS62424-25.0 lbs |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 9630

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## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS
MILITARY

MIL-STD-1884 - Coating, Plasma Spray Deposition.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Printing Service Office, Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B214 - Sieve Analysis of Granular Metal Powders.
ASTM B215 - Sampling Finished Lots of Metal Powders.
ASTM C117 - Materials Finer than 75 um (No. 200) Sieve in Mineral Aggregates by Washing.
ASTM E11 - Wire-Cloth Sieves for Testing Purposes.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.



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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.1 (see 6.3).

3.2 Material. The material shall be as specified herein.

3.2.1 Process. Powder shall be produced by vacuum or inert atmosphere melting process followed by atomizing in an inert atmosphere.

3.2.2 Particle size. The particle size of the powder shall be as specified in table I (see 4.4.3).

TABLE I. Particle size distribution by weight.

| Sieve designation 1/ | | Percentage (%) by weight (maximum) | Weight passed through sieve | Weight retained on sieve |
|---|---|---|---|---|
| Micron | Mesh | | | |
| 75 | 200 | 5 | | X |
| 45 | 325 | 20 | X | |

1/ Sieve designation in accordance with ASTM E11.

3.2.3 Chemical composition. The chemical composition of the material shall conform to the percent by weight for each element as specified in table II (see 4.4.4).

TABLE II. Chemical composition.

| Elements | % by weight | |
|---|---|---|
| | Minimum | Maximum |
| Carbon | -- | 0.02 |
| Chromium | 21.00 | 23.00 |
| Aluminum | 9.00 | 11.00 |
| Yttrium | 0.80 | 1.20 |
| Phosphorus | -- | 0.010 |
| Sulfur | -- | 0.010 |



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TABLE II. Chemical composition - Continued.

| Elements | % by weight | |
|---|---|---|
| | Minimum | Maximum |
| Nitrogen 1/ | -- | 0.020 (200 ppm) 2/ |
| Hydrogen 1/ | -- | 0.005 (50 ppm) |
| Other elements, total 3/ | -- | 0.50 |
| Nickel | Remainder | |

1/ Determination of nitrogen and hydrogen elements shall not be required for acceptance.

2/ ppm = parts per million.

3/ Other elements are the sum percentage of individual elements that may be present not mentioned specifically in the composition. The producer normally shall analyze only for impurities (elements) which are possible to be present because of raw materials or manufacturing processes and which affect the product significantly. It is not implied by this statement that an analysis need be made for each element of the periodic table not mentioned specifically in the composition.

3.2.4 Properties. The powder shall meet the following properties requirements when applied (plasma sprayed) to steel and cobalt, and nickel-base alloys.

3.2.4.1 Bond strength. The bond strength of the plasma-sprayed material shall be not less than 28 megapascals (MPa) [4000 pounds per square inch (psi)] (see 4.4.5).

3.2.4.2 Cup test. The amount of draw before spalling of the coating shall be not less then 7.6 millimeter (mm) (0.300 inch) (see 4.4.5).

3.2.5 Recycled, virgin and reclaimed materials. There are no requirements for the exclusive use of virgin materials. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 4.4.1, 6.4.1 and 6.4.2).

3.3 Unit container. The unit container (see 1.2 thru 1.2.2 and 6.2) shall be capable of holding the following quantities of powder: 0.5, 2.5 and 10 Kg (1, 5 and 25 lbs). The container shall be a sealed moistureproof unit and large enough to permit mixing the powder in it before use. It shall be capable of protecting the powder against damage or contamination until it is ready to be used (see 4.4.1 and 4.4.2). Unit container shall be labeled in accordance with 5.2.



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3.4 Marking. Each container shall be permanently marked to give the following information (see 4.4.2):

Nickel-chromium, aluminum, yttrium powder
PIN
Manufacturer's identification
Quantity
Lot number

3.5 Workmanship. The powder shall be as specified herein and shall be thoroughly blended, uniform in quality and condition, dry and free from contaminants and imperfections which would affect spraying qualities or be detrimental to the performance of coated parts (see 4.4.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 First article. Unless otherwise specified (see 6.2), the supplier of the plasma sprayed parts shall prepare test specimens in accordance with MIL-STD-1884 and in the frequency specified therein for the bond strength and cup tests of 4.4.5.

4.2.1 First article inspection failure. Deficiencies found during, or as a result of, the first article test shall be cause for rejection of the first article until evidence has been provided by the contractor that corrective action has been taken to correct the deficiency. Any deficiency found during, or as a result of the first article test shall be evidence that all items already produced prior to completion of the first article test are



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similarly deficient unless contrary evidence is satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor. The Government will not accept products until the first article test is completed to the satisfaction of the Government.

4.3 Quality Conformance Inspection (QCI). QCI shall include the examination of 4.3.1 and the tests of 4.3.2. Noncompliance with any of the specified requirements in sections 3 and 5 shall be cause for rejection of the sample and the inspection lot (see 6.4).

4.3.1 Examination. Examination shall be conducted on the powder after blending of components Nickel-chromium, aluminum and yttrium.

4.3.1.1 Lot formation. An inspection lot shall consist of all the powder mixed at one time in a blender during an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.3.1.2 Sample. The sample for QCI examination shall be randomly selected from the inspection lot in accordance with table III and defects classified as listed in table IV.

TABLE III. Sampling plan for QCI.

| Inspection lot size | Sample size examination |
|---|---|
| 2 to 8 | 5 |
| 9 to 15 | 5 |
| 16 to 25 | 5 |
| 26 to 50 | 5 |
| 51 to 90 | 7 |
| 91 to 150 | 11 |
| 151 to 280 | 13 |
| 281 to 500 | 16 |
| 501 to 1200 | 19 |
| 1201 to 3200 | 23 |
| 3,201 to 10,000 | 29 |
| 10,001 to 35,000 | 35 |
| 35,001 to 150,000 | 40 |
| 150,001 to 500,000 | 40 |
| 500,000 and over | 40 |



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TABLE IV. Classification of defects.

| Category | Characteristics | Method of inspection |
|---|---|---|
| Critical | None | |
| 101 | Unit container improper size (see 3.3). | Visual |
| 102 | Marking incomplete/improper (see 3.4). | Visual |
| 103 | Workmanship faulty (see 3.5). | Visual |

4.3.2 Tests. Tests shall be conducted on the powder prior to and after blending as specified herein.

4.3.2.1 Lot formation and sample for tests prior to blending all the components. Lot formation and sample for tests shall be in accordance with practice A of ASTM B215. The samples selected shall be subjected to the tests specified in 4.4.3 and 4.4.4. In lieu of testing, the supplier of the powders prior to blending all components shall certify that the tests specified herein have been conducted and met.

4.3.3 QCI failure. Any sample that fails to conform to any specified requirement shall be rejected; any defect/failure (one or more) of the selected sample for the appropriate inspection lot size shall constitute a failure of the entire lot.

4.4 Methods of inspection.

4.4.1 Materials and formulation. Conformance to 3.2, 3.2.1, 3.2.3 and 3.3 shall be determined by inspection of contractor records providing proof or certification that processing and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.4.2 Defects. Conformance to 3.3, 3.4 and 3.5 shall be determined by examination for the defects listed in table IV. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.4.3 Particle size sieve test. The powder shall be tested in accordance with ASTM B214 and C117 to verify conformance to 3.2.2.

4.4.4 Chemical composition test. The powder shall be analyzed to verify conformance to 3.2.3.

4.4.5 Bond strength and cup tests. The bond strength and cup tests shall be conducted in accordance with MIL-STD-1884 to verify conformance to 3.2.4.1 and 3.2.4.2.



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5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging requirements specified by the contracting authority (see 6.2).

5.2 Labeling. In addition to other marking requirements, unit containers and packages shall be conspicuously labeled "CAUTION: Containers contain powdered aluminum. Use approved respirator."

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This powder is primarily intended for use as a plasma spray deposited material providing increased oxidation and hot corrosion resistance.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification
b. Unit container size (see 1.2 thru 1.2.2 and 3.3).
c Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. If first article is required (see 3.1).
e. If responsibility for inspection is other than as specified (see 4.1).
f. Selection of applicable level and packaging requirements (see 5.1).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.2. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely of such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contracting.



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6.4 Definitions.

6.4.1 Definitions of terms used in sampling inspection.

a. Classification of defects. A classification of defects is the enumeration of possible defects of the unit of product classified according to their seriousness. A defect is any nonconformance of the unit of product with specified requirements. Defects will normally be grouped into one or more of the following classes: critical, major and minor defects. Also, defects may be grouped into other classes, or into subclasses within these classes.

b. Critical defects. A critical defect is a defect that judgement and experience indicate would result in hazardous or unsafe conditions for individuals using, maintaining, or depending upon the product, or a defect that judgement and experience indicate is likely to prevent performance of the tactical function of a major end item such as a ship, aircraft, tank, missile, or space vehicle.

c. Critical defective. A critical defective is a unit of product which contains one or more critical defects and may also contain major and/or minor defects.

d. Defective. A defective is a unit of product which contains one or more defects.

e. Formation of lots or batches. The product shall be assembled into identifiable lots, sublots, batches, or in such other manner as may be prescribed (see 1). Each lot or batch shall, as far as is practicable, consist of units of product of a single type, grade, class, size, and composition, manufactured under essentially the same conditions, and at essentially the same time.

f. Lot or batch. The term lot or batch shall mean "inspection lot" or "inspection batch", i.e., a collection of units or product from which a sample is to be drawn and inspected and may differ from a collection of units designated as a lot or batch for other purposes (e.g., production, shipment, etc.).

g. Lot or batch size. The lot or batch size is the number of units of product in a lot or batch.

h. Major defect. A major defect is a defect, other than critical, that is likely to result in failure, or to reduce materially the usability of the unit of product for its intended purpose.

i. Major defective. A major defective is a unit of product which contains one or more major defects, and may also contain minor defects but contains no critical defect.



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j. Minor defect. A minor defect is a defect that is not likely to reduce materially the usability of the unit of product for its intended purpose, or is a departure from established standards having little bearing on the effective use or operation of the unit.

k. Minor defective. A minor defective is a unit of product which contains one or more minor defects but contains no critical or major defect.

l. Presentation of lots or batches. The formation of the lots or batches, lot or batch size, and the manner in which each lot or batch is to be presented and identified by the supplier shall be designated or approved by the responsible authority. As necessary, the supplier shall provide adequate and suitable storage space for each lot or batch, equipment needed for proper identification and presentation, and personnel for all handling of product required for drawing of samples.

m. Representative sampling. When appropriate, the number of units in the sample shall be selected in proportion to the size of sublots or subbatches, or parts of the lot or batch, identified by some rational criterion. When representative sampling is used, the units from each part of the lot or batch shall be selected at random.

n. Sample A sample consists of one or more units of product drawn from a lot or batch, the units of the sample being selected at random without regard to their quality. The number of units or product in the sample is the sample size.

o. Sampling plan. A sampling plan indicates the number of units of product from each lot or batch which are to be inspected (sample size or series of sample sizes) and the criteria for determining the acceptability of the lot or batch (acceptance and rejection numbers).

p. Time of sampling. Samples may be drawn after all the units comprising the lot or batch have been assembled, or samples may be drawn during assembly of the lot or batch.

6.4.2 Recovered materials. "Recovered materials" means materials that have been collected or recovered from solid waste (see 6.4.3).

6.4.3 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining and agricultural operations, and from community activities. It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

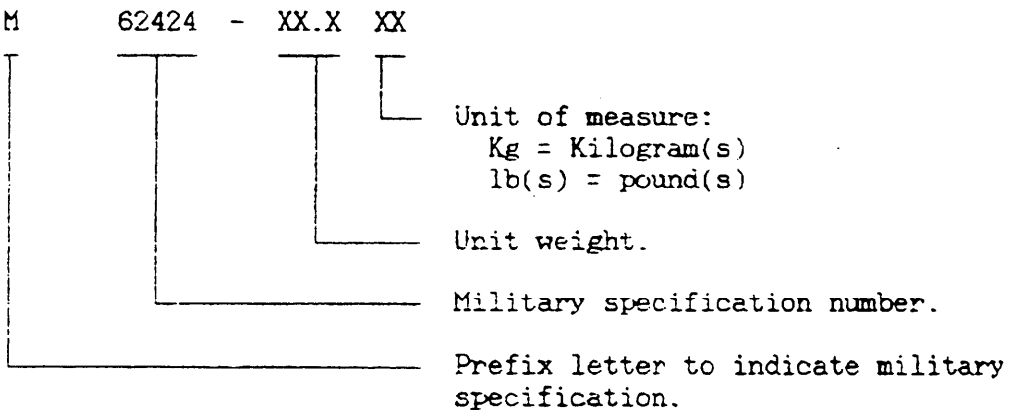


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6.5 Hazardous material. Due to the hazardous nature of this material, the provision of FED-STD-313 apply.

6.6 AMC policy on AQLs/LTPDs. This specification is certified to be in compliance with current Army Material Command (AMC) policy for the elimination of the AQLs/LTPDs (Acceptable Quality Levels/Lot Tolerance Percent Defectives) from military specifications.

6.7 Part or identifying number (PIN) structure. The PINs to be used for materials acquired to this specification are created as follows:




6.8 Subject term (key word) listing.

Bond strength
Chemical composition
Cup test
Particle size
Recovered materials
Sieve size

6.9 Supersession data. This military specification supercedes Textron Lycoming internal specification no. M3960B, dated 26 April 1982.

Custodian:
Army - AT

Preparing activity:
Army - AT

(Project 9630-A022)



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DOCUMENT IDENTIFIER (Number) AND TITLE MIL-P-62424(AT); Powder, Yttrium, Aluminum, Nickel-Chromium (Metric)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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